Serpentine Laser Array

Polarization-Controlled Mode Switching — Streak vs Discrete (1x1 to 6x8 Study)

Red laser dot(s) on a diffuse white LED panel across 22 trials in 5 sessions. **Input polarizer angle selects output mode:** orthogonal (90°) → discrete dots; diamond (45°) → pure streak lines. Streak output is polarized — confirmed by output analyzer extinction. 6x8 expansion planned for collective propagation.

Session 1 — White LED Panel, No Polarizer (01–06)

File	Dur.	Description
01_baseline_single_dot.mp4	~48s	Baseline — single localized mode. One stationary dot at lower-center. No lattice propagation; edge do
02_dot_appears_bottom.mp4	~48s	Mode re-entry at edge. Dot reappears at bottom edge with slight drift. Coupling path still minimal.
03_dots_cluster_scatter.mp4	~47s	Discrete satellite cluster onset. Multiple dots in loose irregular cluster. Transition from single mode tow
04_dots_arc_pattern.mp4	~43s	Angular preference emerging. Dots in curved arc — non-circular symmetry indicates serpentine geom
05_streaks_vertical.mp4	~35s	Continuous fringe to streak transition. Dots elongate into vertical streaks; partial fringe coupling along
06_streaks_full_panel.mp4	~34s	Broadest local mode extent. Multiple streaks across panel — closest to collective behavior in 1x1, but

Session 2 — Dark Background / UV Suppression (07–10)

File	Dur.	Description
07_discrete_cluster_dark.mp4	~41s	Discrete satellite cluster confirmed — dark field. Sharply separated point emitters; non-circular asymm
08_cluster_to_streaks_dark.mp4	~33s	Cluster to streak transition — dark. Same preferred axes as Session 1, confirming structural optical pa
09_scattered_dots_dark.mp4	~38s	Wide-field discrete scatter — dark. Dots do not blur or merge; consistent with localized mode jumping
10_uv_suppression_red_on_blue.mp4	~36s	UV suppression trial. Background shifts to blue/violet. Red satellites remain visible as distinct warm en

Session 3 — Black Paper Backup (11–15)

File	Dur.	Description
11_blackpaper_single_streak_onset.mp4	~40s	Single dot with streak onset — black paper. Fringe suppression not a white-panel artifact.
12_blackpaper_fringe_suppressed_dot.mp4	~34s	Fringe fully suppressed — black paper. Single dot, no diffuse halo. Surface-independent fringe suppre
13_blackpaper_satellite_cluster_cross.mp4	~35s	Satellite cluster — cross geometry. Discrete dots in plus pattern; angular preference consistent with S
14_blackpaper_fringe_null_dark.mp4	~31s	Near-complete fringe null — black paper. Fringe suppression reaches near-zero floor independent of
15_blackpaper_wide_discrete_scatter.mp4	~28s	Wide-field discrete scatter — black paper. Pure point-like emitters in broad arc; matches LED panel be

Session 4 — Orthogonal Polarizer at Input (16–20)

Orthogonal (90°) polarizer at input → discrete dot mode even in 1x1 array. Array size is not the primary driver.

File	Dur.	Description
16_polarizer_orthogonal_single_dot.mp4	~29s	Orthogonal polarizer baseline. Single dot, no fringe lines. Polarization state controls fringe onset.
17_polarizer_orthogonal_fringe_null.mp4	~26s	Fringe fully nulled — orthogonal. Fringe lines are polarization-dependent, not a geometry artifact.
18_polarizer_orthogonal_dot_stable.mp4	~27s	Stable isolated dot — orthogonal. System locked in single localized state.
19_polarizer_orthogonal_discrete_onset.mp4	~22s	Discrete satellite onset — 1x1, orthogonal polarizer. Confirms discrete mode generation is polarization
20_polarizer_orthogonal_discrete_confirmed.mp4	~24s	Discrete mode confirmed — polarizer-induced, geometry-independent. Orthogonal input is sufficient fo

Session 5 — Diamond Polarizer at Input / Output Rotation (21–22)

Diamond (45°) polarizer at input → pure streak mode. Output polarizer rotation causes extinction then recovery, confirming streaks carry defined polarization.

File	Dur.	Description
21_polarizer_diamond_input_streak_mode.mp4	~6s	Diamond polarizer at input — pure streak mode. Input at 45° produces entirely streak output with no d
22_polarizer_output_rotation_streak_suppress_recover.mp4	~30s	Output polarizer rotation — streak suppression and recovery. Streaks temporarily disappear at extinct

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Technical Notes

- **Camera:** 1920x1080 @ ~30 fps (original)
- **Panel:** Diffuse white surface (LED panel / acrylic diffuser)
- **Laser:** Red (~650 nm), single-point source
- **Compressed:** 960x540, H.264 CRF 26-28, 15 fps — optimized for GitHub (<2 MB/file)

Analysis

Based on GPT analysis of the footage.

The 1x1 serpentine behaves as **localized mode switching**: no global lattice propagation, short coupling paths, edge acts as the entire system. Pattern jumps between discrete states rather than flowing continuously.

Angular Preference

Satellite clusters are not circularly symmetric — the same directional set reappears across all sessions, indicating structural **optical path selection**, not passive scattering.

Input Polarizer as Mode Selector

Orthogonal (90°) input → discrete dot satellites. Diamond (45°) input → pure streak lines. Output polarizer extinction confirms streak output is **coherently polarized**. Mode selection is therefore a direct function of input polarization angle.

Expected Changes with 6x8 Expansion

- Fringe plumes will extend and connect spatially
- Satellite activations will propagate in sequence across the array
- The 'frozen dot' appearance is expected to diminish significantly

Summary Assessment (Sessions 1–5)

Item	Status
Continuous fringe to discrete satellite transition	Partially observed ✓
Behavior confined to local mode level	Confirmed ✓
Discrete character confirmed under dark field	Confirmed (S2) ✓
Fringe suppression surface-independent	Confirmed (S3) ✓
Red satellite survives UV suppression	Confirmed (video 10) ✓
Orthogonal polarizer input → discrete dots, 1x1 sufficient	**Confirmed (S4) ✓**
Diamond polarizer input → pure streak mode	**Confirmed (video 21) ✓**
Streak output carries defined polarization state	Confirmed (video 22) ✓
Large-array collective effect	Pending — 6x8 required ✗

Conclusion: Input polarizer angle is the primary mode selector — orthogonal (90°) produces discrete dots; diamond (45°) produces pure streak lines. Streak output is coherently polarized, confirmed by analyzer extinction. 6x8 expansion remains necessary for large-scale collective propagation.